

UNIT 3: Statistical philosophy

Overview

- ▶ We use statistics to confirm effects, estimate parameters, and predict outcomes
- ▶ I think smoking is bad for lung capacity in children
 - ▶ *Confirmation:* Children who smoke have lower exhalation volumes (ev)
 - ▶ *Estimation:* The average non-smoker has an ev 1.1–1.3 times higher than a smoker
 - ▶ *Prediction:* A 14-year-old male smoker will have an ev of 2.6–3.0 L/s

Some definitions

- ▶ a **variable** is something that we measure
 - ▶ flower species, rodent weight, provincial area
- ▶ a **statistic** is something that we calculate from our variables
 - ▶ mean weight, median area, correlation between weight and leg length
- ▶ a **parameter** is something we estimate using a statistical model
 - ▶ the difference between soil wetness surrounding two different types of plants
 - ▶ parameters are estimated with uncertainty

Example: Vitamin A

- ▶ We compare health indicators of children treated or not treated with vitamin A supplements
 - ▶ *Estimate*: how much healthier (or less healthy) are the treated children on average?
 - ▶ *Confirmation*: are we sure that the supplements are helping (or hurting)?
 - ▶ *Range of estimates*: how much do we think the supplement is helping?

P values and confidence intervals

- ▶ We use *P values* to say how sure we are that we have seen a positive effect
- ▶ We use *confidence intervals* to say what we think is going on (with a certain level of confidence)
- ▶ P values are *over-rated*
- ▶ *Never* use a high P value as (direct) evidence for anything, e.g.:
 - ▶ that an effect is small
 - ▶ that two quantities are similar
- ▶ What does a P value actually measure?

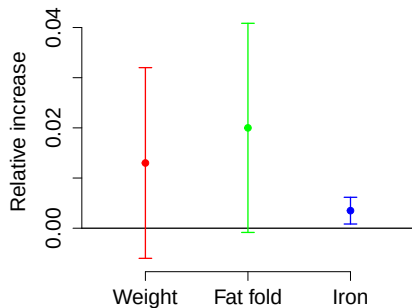
What do we hope to learn?

- ▶ Is vitamin A good for these children?
- ▶ How sure are we?
- ▶ How good do we think it is?
- ▶ How sure are we about that?

P values

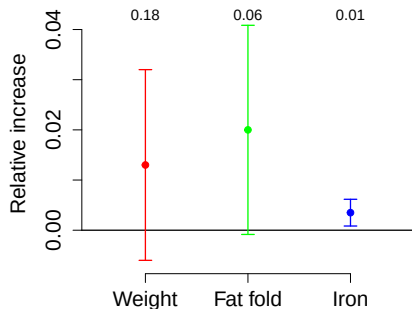
- ▶ What is a P value?
 - ▶ * The probability of finding a result at least this “extreme” under the null hypothesis
- ▶ What does it mean if I find a “significant P value” for some effect in this experiment?
 - ▶ * A difference this big would be unlikely under the null hypothesis
 - ▶ * So what! I already know vitamin A has strong effects on metabolism
 - ▶ It's unlikely this group of children is healthier just at random
- ▶ If I'm certain that the null hypothesis is false, why do I test it anyway?
 - ▶ * NOANS
 - ▶ *Because I want to know if I'm confident about the direction!*

Confidence intervals



- ▶ What do these results mean?
- ▶ Which are significant?

Confidence intervals and P values



- ▶ A high P value means we can't see the *sign* of the effect clearly
- ▶ A low P value means we can
- ▶ What would happen if we could measure many more children?

Outline

Frequentist paradigm

P value rant

Types of error

Frequentist paradigm

- ▶ Choose an allowable mistake rate, α
 - ▶ * Almost always 0.05
 - ▶ *Because Fisher made that up as an example!*
- ▶ The paradigm is: our probability of being wrong should be $\leq \alpha$
- ▶ If we do a lot of studies (correctly), we won't wrong more than α of the time
- ▶ An approach that correctly limits the probability of being wrong is **valid**

Frequentist P-value logic

- ▶ I define a P value as the probability under the null of getting a value at least as extreme as my observation
- ▶ Hypothetically (or counter-factually) if the null is true, what will my P value be?
 - ▶ * 5% of the time it will be $\leq 5\%$
 - ▶ * α of the time it will be $\leq \alpha$.
 - ▶ * This meets my frequentist requirement

Comparing different tails

- ▶ Let's say I have 5 outcomes and my null hypothesis is that they are all equally likely.
 - ▶ I observe 10% (instead of 20%) of the key outcome
 - ▶ Is that more or less “extreme” than observing 35%?
 - ▶ * Not clear! 10% is getting close to the boundary of 0.
- ▶ I start with 10g of calcium in a mesocosm and expect that to stay the same?
 - ▶ What lower value is as extreme as 20g?
 - ▶ * 0g, or 5g, or something in between

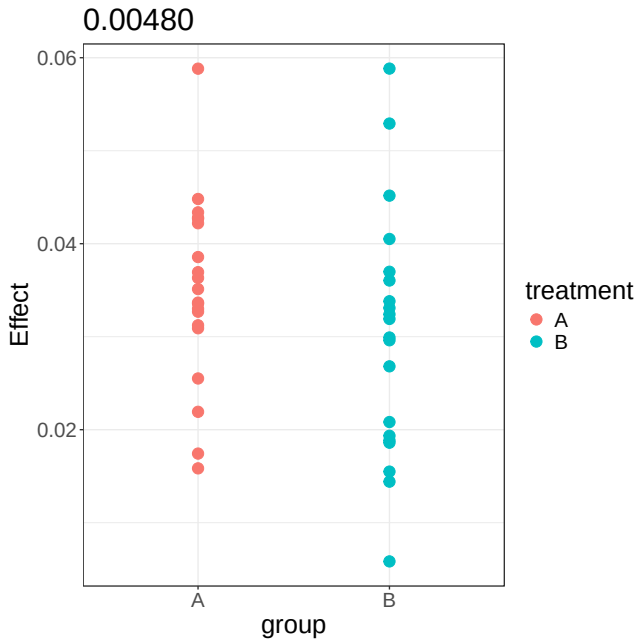
α spending

- ▶ JD: In simple situations it's usually either equivalent or better to *not* compare different directions
 - ▶ Just ask what's the probability of seeing something this extreme or more in this direction —
 - ▶ * But you don't want to double your probability of being wrong
 - ▶ * You should divide your cutoff by 2 (spend half of your α on each direction)
 - ▶ * "Spend" half of your α on each of the two directions
 - ▶ * It is more common (but less beautiful) to multiply the P value by 2.

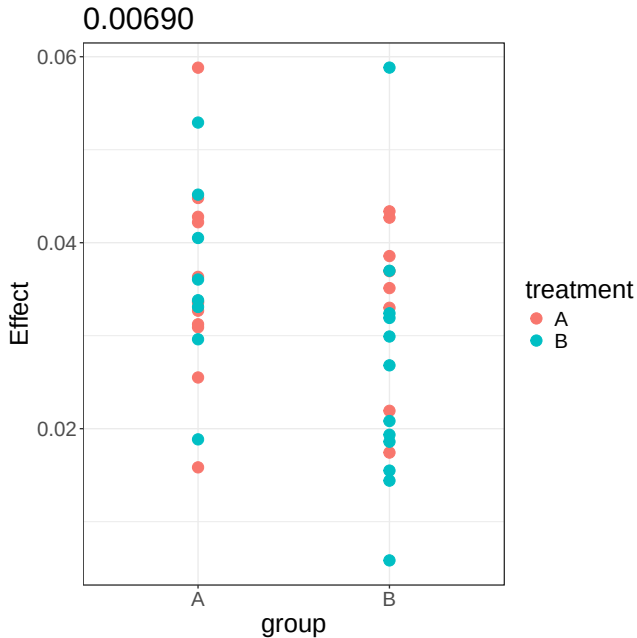
The most basic frequentist test

- ▶ What would have happened if our treatment had no effect at all?
- ▶ How does that compare to what we saw?
- ▶ What's the simplest way to do this?
 - ▶ * Just relabel the children!

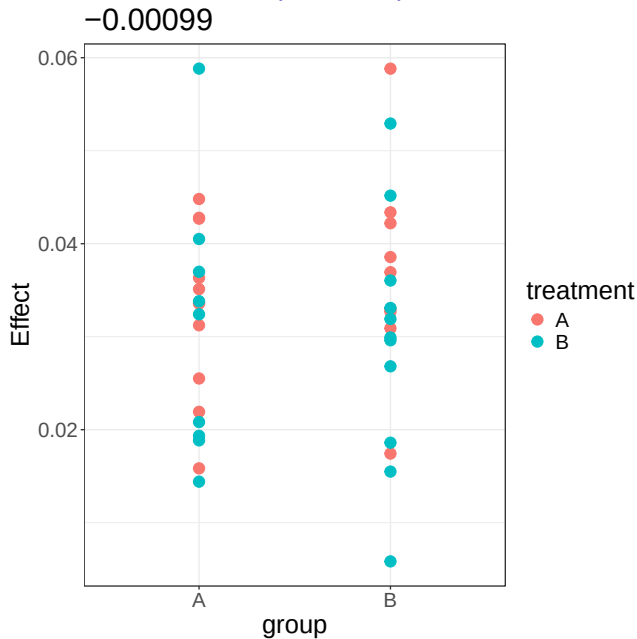
Vitamin results



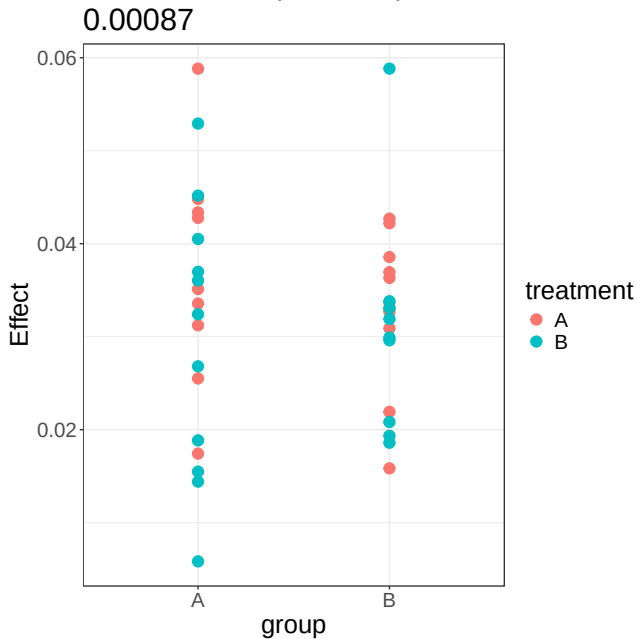
Scrambled measurements



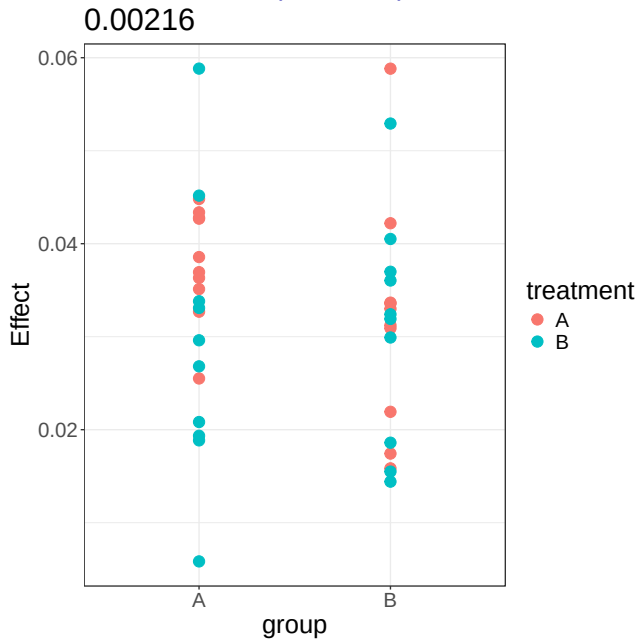
Scrambled measurements (present)



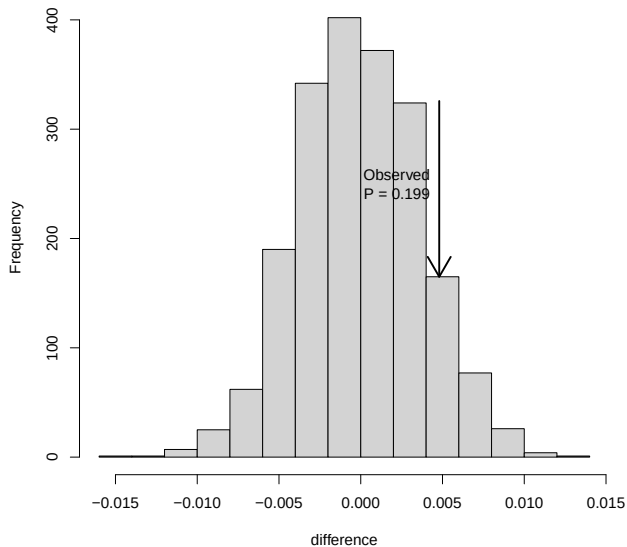
Scrambled measurements (present)



Scrambled measurements (present)

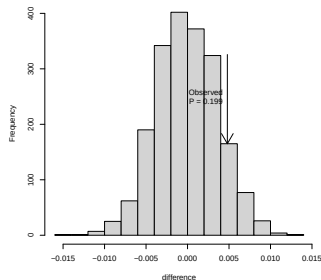


The null distribution



Interpretation

- ▶ 10% of random trials had an effect as bigger or bigger than the observed one
- ▶ Compare $P = 0.10$ to $\alpha = 0.05$
 - ▶ Or one-sided $P = 0.05$ to $\alpha = 0.025$
- ▶ * We don't want to conclude the effect was positive
 - ▶ * Because then our chance of making a mistake would be too big
 - ▶ * We definitely don't want to conclude that it was not positive



Practical frequentism

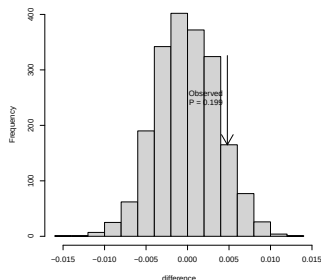
- ▶ Making assumptions allows us to do much more **powerful** statistics
 - ▶ * More likely to find real effects when they exist
- ▶ But assumptions are generally not exactly correct
 - ▶ Error rate could be higher or lower than you think
- ▶ There's also a tradition of accepting tests that are *asymptotically valid*
 - ▶ * Error rate is $\leq \alpha$ with the additional assumption that there is “enough” data
 - ▶ * Normal approximation to binomial tests
 - ▶ * ANOVA for non-normal observations

Practical frequentism

- ▶ The bedrock of frequentism is testing the null hypothesis
 - ▶ * In the hypothetical where there's nothing to detect, we want our method to detect it $\leq \alpha$ of the time
- ▶ We also want our confidence intervals to be valid
 - ▶ * The parameter we're estimating should fall outside our confidence interval $\leq \alpha$ of the time
- ▶ But this is harder
 - ▶ And people are less strict about it

Permutation test assumptions

- ▶ What assumptions did we make for our permutation test?
 - ▶ * Identical populations
 - ▶ * Achieved by randomization
 - ▶ * Independent sampling
 - ▶ * What if we gave siblings the same treatment?



Thinking about populations

- ▶ All the children in the village?
- ▶ A larger group of the kind of children you would hypothetically get under these circumstances?

Confidence interval logic

- ▶ To reject the null hypothesis, we ask:
 - ▶ assuming the null is true, what is the probability under the null of getting a value at least as extreme as my observation?
- ▶ To reject a value from our confidence interval, we ask:
 - ▶ assuming the value is true, what is the probability under the null of getting a value at least as extreme as my observation?
- ▶ Result:
 - ▶ * We reject the (unknown) true value $\leq \alpha$ of the time
 - ▶ * If all of our assumptions are correct

Confidence intervals vs. P values

- ▶ CIs are more useful
- ▶ They also require more assumptions
 - ▶ If effects are absent, you don't need any assumptions about them
 - ▶ To assume that treated children grow better, we need assumptions about the relationship between distributions

Paradigm summary

- ▶ We want to “fail to reject” the null hypothesis at least $1 - \alpha$ of the time, under the assumption that it's true
 - ▶ * We can do this by only rejecting outcomes that are unlikely under the null hypothesis
 - ▶ * Using our P value definition
- ▶ We (usually) do this because we want to test confidence in the *sign* of an effect or difference
 - ▶ * The probability of making a sign error will be less than the probability of a “false positive”
 - ▶ * If the true answer is negative, the probability of seeing our positive observation will be less than it would be under the null hypothesis

Outline

Frequentist paradigm

P value rant

Types of error

Low P values

- ▶ If I have a low P value I can see something clearly
- ▶ But it's usually better to focus on what I see than the P value



High P values

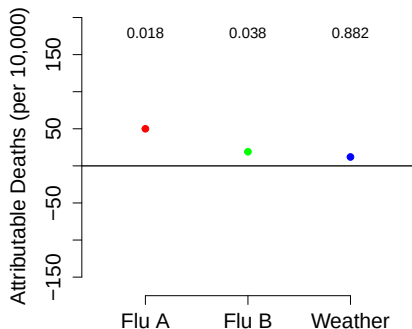
- ▶ If I have a high P value, there is something I *don't* see clearly
- ▶ It *may be* because this effect is small
- ▶ High P values should *not* be used to advance *any* scientific conclusion



What causes high P values?

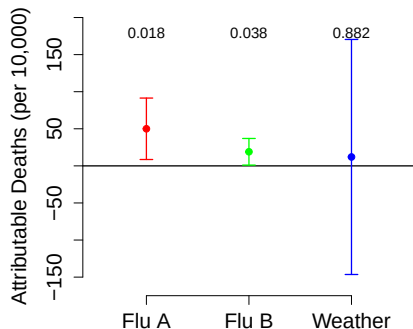
- ▶ Small differences
- ▶ Less data
- ▶ More noise
- ▶ An inappropriate model
- ▶ Less model resolution
- ▶ A lower P value means that your evidence for difference is better
- ▶ A higher P value means that your evidence for similarity is better – or worse!

Annualized flu deaths



- ▶ Look at deaths on an annual time scale
- ▶ Why is weather not causing deaths at this time scale?
- ▶ * Maybe those people would have died anyway later in the same winter

... with confidence intervals



- ▶ **Never** say: A is significant and B isn't, so $A > B$
- ▶ **Instead:** Construct a statistic for the hypothesis $A > B$
 - ▶ May be difficult

What do P values measure?



- ▶ * Clarity!
- ▶ * We should call it that
 - ▶ * Instead of significance

Rant summary

- ▶ We don't accept the null hypothesis
 - ▶ Statistics isn't magic!
- ▶ It is easy to make this mistake
 - ▶ and the word “significant” makes this problem worse
- ▶ A lot of advice you will see is just wrong
 - ▶ e.g., drop non-significant terms

Outline

Frequentist paradigm

P value rant

Types of error

Example: vaccines

- ▶ Is MMR vaccination associated with autism outcomes?
 - ▶ * Yes!
- ▶ Is the association positive or negative?
 - ▶ * Nobody knows!
- ▶ What does this tell us?
 - ▶ * Not to trust anyone who says they do know
- ▶ What would be a better question?
 - ▶ * Is the correlation big or small?
 - ▶ * There's evidence that it's very small
 - ▶ * And no good evidence why the link should exist

Statistical error types

- ▶ Type I (*False positive*:) concluding there is an effect when there isn't one
 - ▶ This doesn't happen in biology. There is always an effect.
 - ▶ But it is useful to think about as a counter-factual
- ▶ Type II (*False negative*:) concluding there is no effect when there really is
 - ▶ This *should* never happen, because we should never conclude there is no effect
 - ▶ We should think about “failure to find” and “power to find”
 - ▶ Later in the term

Doing Science

- ▶ Once you collect data:
- ▶ You should never make a false-negative error.
 - ▶ * Never accept the null
- ▶ Or a false-positive error
 - ▶ * If you never believe the null is exactly true, the idea of a false positive doesn't make sense
- ▶ *Sign error*: if I think an effect is positive, when it's really negative (or vice versa)
- ▶ *Magnitude error*: if I think an effect is small, when it's really large (or vice versa)
- ▶ Confidence intervals aid clear thinking

Small effects

- ▶ We shouldn't build statistical confidence that something is small by failing to see it clearly
- ▶ We must instead see clearly that it is small
- ▶ This means we need a standard (our cutoff) for what we mean by small

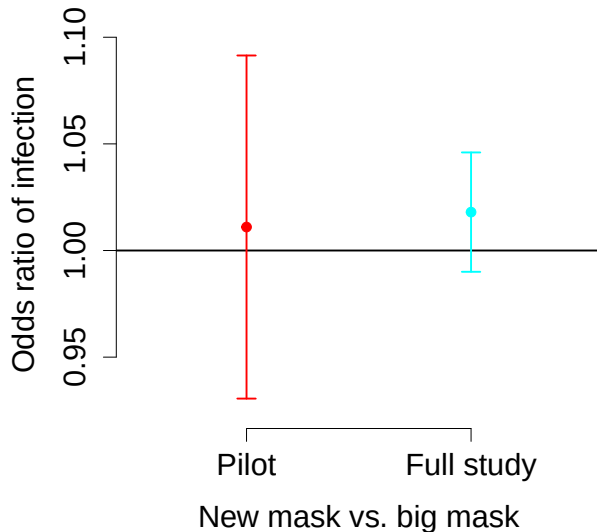
Flu masks (present)



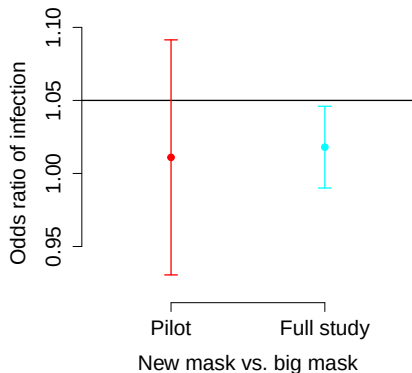
Flu mask example

- ▶ People who work in respiratory clinics sometimes have to wear bulky, uncomfortable, expensive masks
- ▶ They would like to switch to simpler masks, if those will do the job
- ▶ How can this be tested statistically? We don't want the masks to be “different”.
 - ▶ We need to decide what we mean by different in this case!
 - ▶ They're not the same, so how close is close enough?

Traditional statistics



Non-inferiority approach



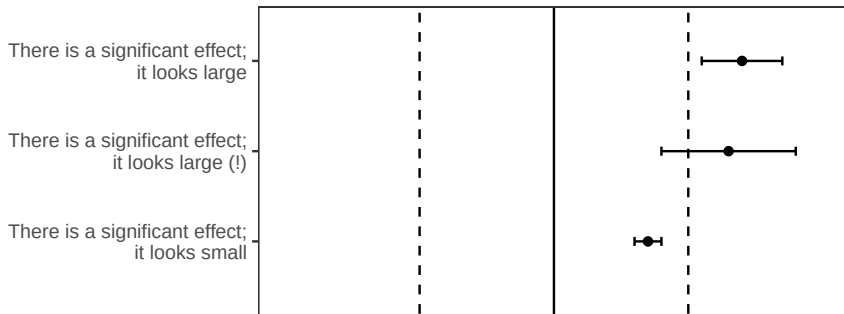
- ▶ Are we confident the new mask is “good enough”?
- ▶ There is no substitute for picking a cutoff
- ▶ Can be controversial

Is your effect big?

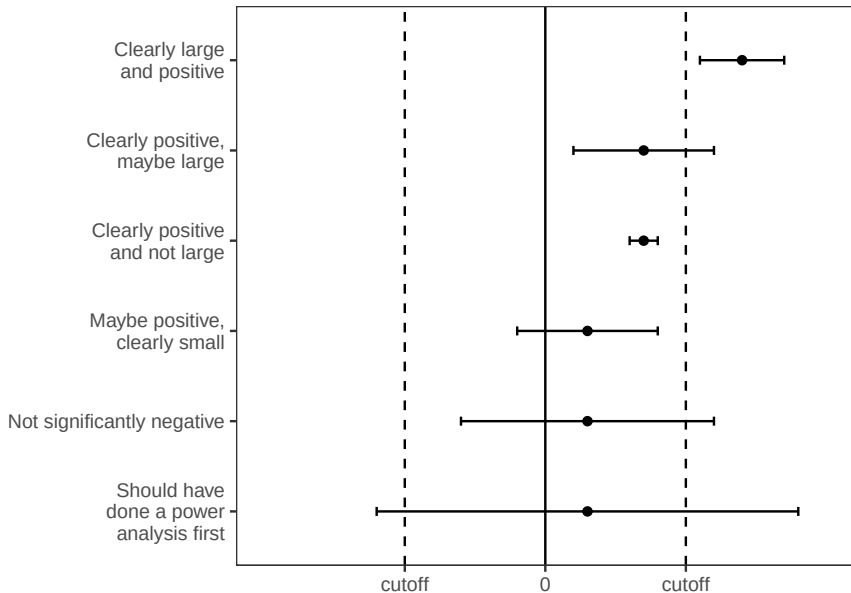
- ▶ Scientists in many fields resist picking effect sizes
- ▶ They want statistics to do it for them
 - ▶ ... but statistics is not magic
 - ▶ scientific judgment is required
- ▶ It would be great if we could use “significant” to mean “biologically important”
 - ▶ * Would probably create too much confusion

Summary

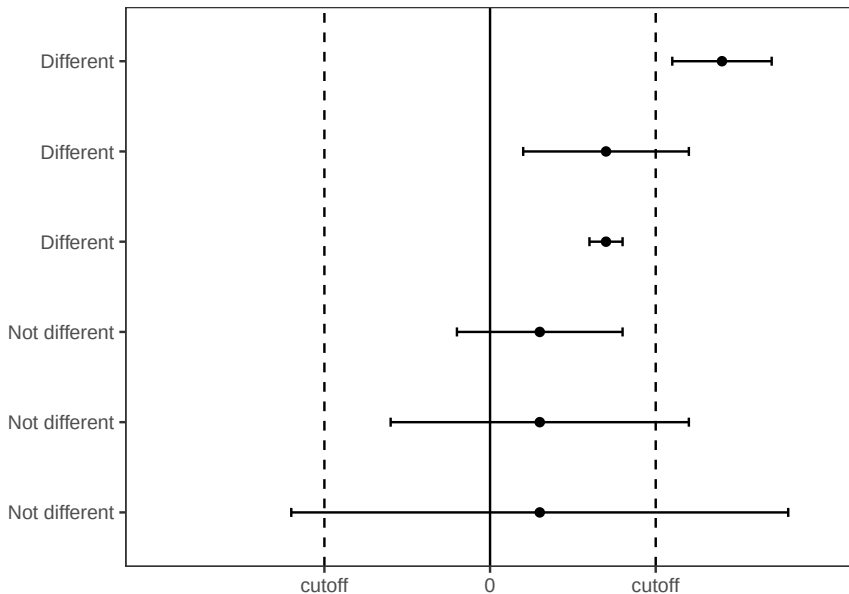
- ▶ Not seeing an effect clearly should not be used as evidence that the effect is small, or unimportant
- ▶ Your confidence intervals are what tell you how confident you are that an effect is below a certain size
- ▶ To say that something is bigger than something else, you have to compare them
 - ▶ * The difference between “significant” and “not significant” is not necessarily significant!
 - ▶ * Influenza example Confidence intervals



Confidence intervals



Confidence intervals



Cutoffs

- ▶ How do I know what is a large effect?
- ▶ This can't be done by statistics
 - ▶ * Not magic!
- ▶ Must be scientific
 - ▶ It's often hard, and people avoid it
 - ▶ This leads to sloppy thinking